

Problem Set 32

Substitution

1

Read pages 787 - 796 in Gottlieb's Textbook. If you'd like a little extra practice, Exercise 25.2 on page 796 gives you some hints without giving all of the details. You can also try the problems on the in-class worksheet that you didn't get to in class (solutions are posted on the course website).

Evaluate the following integrals using the given substitution. (Can you see why the given substitution is a good choice?)

Check your answers to (a) and (c) by differentiating them.

(a) $\int x \sin(x^2) dx$, substituting $u = x^2$

Solution

$u = x^2$ is a good substitution since x^2 is the inner function in the composition $\sin(x^2)$, and we see something related to $du = 2x dx$ in the integral, namely $\frac{1}{2} du = x dx$. Then,

$$\begin{aligned} \int x \sin(x^2) dx &= \int \sin(u) \cdot \frac{1}{2} du \\ &= -\frac{1}{2} \cos u + C \\ &= \boxed{-\frac{1}{2} \cos(x^2) + C} \end{aligned}$$

Let's check:

$$\begin{aligned} \frac{d}{dx} \left(-\frac{1}{2} \cdot \cos(x^2) + C \right) &= -\frac{1}{2} \cdot (-\sin(x^2) \cdot 2x) \\ &= x \sin(x^2) \end{aligned}$$

(b) $\int_0^1 \frac{1}{(4x+1)^3} dx$, substituting $u = 4x+1$

Solution

$u = 4x+1$ is a good substitution since $4x+1$ is the inner function in the composition $(4x+1)^3$. With this substitution, $du = 4 dx$, so $\frac{1}{4} du = dx$. Since x goes from 0 to 1, $u = 4x+1$ goes from

$4(0) + 1 = 1$ to $4(1) + 1 = 5$, so we can rewrite the original integral as

$$\begin{aligned}\int_0^1 \frac{1}{(4x+1)^3} dx &= \int_1^5 \frac{1}{u^3} \cdot \frac{1}{4} du \\ &= \frac{1}{4} \int_1^5 u^{-3} du \\ &= \frac{1}{4} \left(-\frac{1}{2} u^{-2} \Big|_1^5 \right) \\ &= -\frac{1}{8} \cdot \frac{1}{u^2} \Big|_1^5 \\ &= -\frac{1}{8} \left(\frac{1}{25} - 1 \right) \\ &= -\frac{1}{8} \left(-\frac{24}{25} \right) \\ &= \boxed{\frac{3}{25}}\end{aligned}$$

Note on notation: If you prefer not to change the bounds of the integral to be in terms of u , your first line should instead read

$$\int_0^1 \frac{1}{(4x+1)^3} dx = \int_{x=0}^{x=1} \frac{1}{u^3} \cdot \frac{1}{4} du.$$

It is **not** correct to write

$$\int_0^1 \frac{1}{(4x+1)^3} dx = \int_0^1 \frac{1}{u^3} \cdot \frac{1}{4} du.$$

(The latter notation says that u is going from 0 to 1, which is not true.)

(c) $\int \frac{x}{x^2+9} dx$, substituting $u = x^2 + 9$

Solution

$u = x^2 + 9$ looks like a reasonable substitution since it's in the denominator and we see something related to $du = 2x dx$ in the integral, namely $\frac{1}{2} du = x dx$. Then,

$$\begin{aligned}\int \frac{x}{x^2+9} dx &= \int \frac{1}{u} \left(\frac{1}{2} du \right) \\ &= \frac{1}{2} \int \frac{1}{u} du \\ &= \frac{1}{2} \ln |u| + C \\ &= \boxed{\frac{1}{2} \ln |x^2 + 9| + C}\end{aligned}$$

Let's check. First, notice that $x^2 + 9$ is always positive, so $|x^2 + 9|$ is the same as $x^2 + 9$. Therefore, another way of writing our answer is $\frac{1}{2} \ln(x^2 + 9) + C$:

$$\frac{d}{dx} \left(\frac{1}{2} \ln(x^2 + 9) \right) = \frac{1}{2} \cdot \frac{1}{x^2 + 9} \cdot 2x = \frac{x}{x^2 + 9}$$

(d) $\int_0^{\ln 4} e^x \sqrt{3e^x + 13} dx$, substituting $u = 3e^x + 13$

Solution

$u = 3e^x + 13$ looks like a good substitution since it's the inner function in the composition $\sqrt{3e^x + 13}$, and we see something related to $du = 3e^x dx$ in the integral, namely $\frac{du}{3} = e^x dx$. Since x goes from 0 to $\ln 4$, u goes from $3e^0 + 13 = 16$ to $3e^{\ln 4} + 13 = 3 \cdot 4 + 13 = 25$:

$$\begin{aligned} \int_0^{\ln 4} e^x \sqrt{3e^x + 13} dx &= \int_{16}^{25} \sqrt{u} \frac{du}{3} \\ &= \frac{1}{3} \int_{16}^{25} u^{1/2} du \\ &= \frac{1}{3} \cdot \frac{2}{3} u^{3/2} \Big|_{16}^{25} \\ &= \frac{2}{9} \cdot 25^{3/2} - \frac{2}{9} \cdot 16^{3/2} \\ &= \frac{2}{9} \cdot 5^3 - \frac{2}{9} \cdot 4^3 \\ &= \boxed{\frac{122}{9}} \end{aligned}$$

2

In one of the following integrals, the substitution $u = e^{2x} + 1$ works well; in the other, the substitution $u = e^x$ does (it's up to you to decide which is which). Evaluate both integrals, and check your answers by differentiating them.

(a) $\int \frac{2e^x}{e^{2x} + 1} dx$

Solution

Here, it works well to substitute $u = e^x$, since we also see $du = e^x dx$ in the integral. Note that e^{2x} can be rewritten as $e^x \cdot e^x$, or $(e^x)^2$:

$$\begin{aligned} \int \frac{2e^x}{(e^x)^2 + 1} dx &= \int \frac{2}{u^2 + 1} du \\ &= 2 \arctan u + C \\ &= \boxed{2 \arctan(e^x) + C} \end{aligned}$$

Let's check:

$$\frac{d}{dx} \left(2 \arctan \left(\boxed{e^x} \right) \right) = \frac{2}{(e^x)^2 + 1} \cdot e^x$$

(b) $\int \frac{e^{2x}}{e^{2x} + 1} dx$

Solution

Here, it works better to substitute $u = e^{2x} + 1$ because we see something related to $du = 2e^{2x} dx$

in the integral, namely $\frac{du}{2} = e^{2x} dx$. So,

$$\begin{aligned}\int \frac{e^{2x}}{e^{2x} + 1} dx &= \int \frac{1}{u} \cdot \frac{du}{2} \\ &= \frac{1}{2} \int \frac{1}{u} du \\ &= \frac{1}{2} \ln |u| + C \\ &= \boxed{\frac{1}{2} \ln |e^{2x} + 1| + C}\end{aligned}$$

Note that $e^{2x} + 1$ is always positive, so we can also rewrite this as simply $\frac{1}{2} \ln |e^{2x} + 1| + C$. Let's check that this is right by differentiating it:

$$\frac{d}{dx} \left(\frac{1}{2} \ln \left(\boxed{e^{2x} + 1} \right) \right) = \frac{1}{2} \cdot \frac{1}{e^{2x} + 1} \cdot 2e^{2x} = \frac{e^{2x}}{e^{2x} + 1}$$

3

(parts of 25.2.3, 25.2.4) Evaluate the following integrals. (It may take you a few tries to pick a useful u ; that's normal when you're first learning substitution!)

(a) $\int \frac{1}{\sqrt{2x+1}} dx$

Solution

It looks like a good substitution is $u = 2x + 1$, since $2x + 1$ is the inner function in the composition $\sqrt{2x + 1}$. With this substitution, $du = 2 dx$, so $\frac{1}{2} du = dx$. Then,

$$\begin{aligned}\int \frac{1}{\sqrt{2x+1}} dx &= \int \frac{1}{\sqrt{u}} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \int u^{-1/2} du \\ &= u^{1/2} + C \\ &= \boxed{\sqrt{2x+1} + C}\end{aligned}$$

(b) $\int \frac{1}{2x+1} dx$

Solution

There's no composition here, but it could be helpful to substitute for the denominator, so let's try $u = 2x + 1$. With this substitution, $du = 2 dx$, so $\frac{1}{2} du = dx$. Then,

$$\begin{aligned}\int \frac{1}{2x+1} dx &= \int \frac{1}{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \ln |u| + C \\ &= \boxed{\frac{1}{2} \ln |2x+1| + C}\end{aligned}$$

(c) $\int x\sqrt{2x^2 + 1} \, dx$

Solution

It looks like a good substitution is $u = 2x^2 + 1$, since $2x^2 + 1$ is the inner function in the composition $\sqrt{2x^2 + 1}$, and we see something related to $du = 4x \, dx$ in the integral, namely $\frac{du}{4} = x \, dx$:

$$\begin{aligned} \int x\sqrt{2x^2 + 1} \, dx &= \int \sqrt{u} \frac{du}{4} \\ &= \frac{1}{4} \int u^{1/2} \, du \\ &= \frac{1}{4} \cdot \frac{2}{3} u^{3/2} + C \\ &= \boxed{\frac{1}{6} (2x^2 + 1)^{3/2} + C} \end{aligned}$$

(d) $\int \frac{x}{\sqrt{2x^2 + 1}} \, dx$

Solution

It looks like a good substitution is $u = 2x^2 + 1$, since $2x^2 + 1$ is the inner function in the composition $\sqrt{2x^2 + 1}$, and we see something related to $du = 4x \, dx$ in the integral, namely $\frac{du}{4} = x \, dx$:

$$\begin{aligned} \int \frac{x}{\sqrt{2x^2 + 1}} \, dx &= \int \frac{1}{\sqrt{u}} \cdot \frac{du}{4} \\ &= \frac{1}{4} \int u^{-1/2} \, du \\ &= \frac{1}{4} \cdot 2u^{1/2} + C \\ &= \boxed{\frac{1}{2} \sqrt{2x^2 + 1} + C} \end{aligned}$$

4

(parts of 25.2.4, 25.2.6, 25.2.18) Evaluate the following integrals. Pay attention to whether you are working on an indefinite integral or a definite integral and whether your final answer makes sense.

(a) $\int_{1/4}^1 \frac{\cos(\sqrt{x})}{\sqrt{x}} \, dx$

Solution

A promising substitution is $u = \sqrt{x}$, since it's the inner function in the composition $\cos(\sqrt{x})$. Then, $du = \frac{1}{2}x^{-1/2} \, dx = \frac{1}{2\sqrt{x}} \, dx$, which is good because we see something related to this in the

integral: $\frac{1}{\sqrt{x}} dx = 2 du$. As x goes from $\frac{1}{4}$ to 1, $u = \sqrt{x}$ goes from $\sqrt{\frac{1}{4}} = \frac{1}{2}$ to $\sqrt{1} = 1$. So,

$$\begin{aligned} \int_{1/4}^1 \frac{\cos(\sqrt{x})}{\sqrt{x}} dx &= \int_{1/2}^1 \cos u \cdot 2 du \\ &= 2 \int_{1/2}^1 \cos u du \\ &= 2 \sin u \Big|_{1/2}^1 \\ &= \boxed{2 \sin 1 - 2 \sin \frac{1}{2}} \end{aligned}$$

(b) $\int \sqrt{\cos x} \sin x dx$

Solution

It looks like a good substitution is $u = \cos x$, since $\cos x$ is the inner function in the composition $\sqrt{\cos x}$; in addition, we see something related to $du = -\sin x dx$ in the integral, namely $-du = \sin x dx$. Then,

$$\begin{aligned} \int \sqrt{\cos x} \sin x dx &= \int \sqrt{u} (-du) \\ &= - \int u^{1/2} du \\ &= -\frac{2}{3} u^{3/2} + C \\ &= \boxed{-\frac{2}{3} (\cos x)^{3/2} + C} \end{aligned}$$

(c) $\int \tan x dx$

Solution

Let's first rewrite $\tan x$ as $\frac{\sin x}{\cos x}$, so that we are trying to evaluate $\int \frac{\sin x}{\cos x} dx$. It looks like $u = \cos x$ is a promising substitution because we also see something related to $du = -\sin x dx$, namely $-du = \sin x dx$:

$$\begin{aligned} \int \frac{\sin x}{\cos x} dx &= \int \frac{1}{u} (-du) \\ &= - \int \frac{1}{u} du \\ &= -\ln |u| + C \\ &= \boxed{-\ln |\cos x| + C} \end{aligned}$$

(d) $\int_0^{\ln 5} \frac{3e^x}{\sqrt{e^x + 4}} dx$

Solution

It looks like a good substitution is $u = e^x + 4$, since $e^x + 4$ is the inner function in the composition $\sqrt{e^x + 4}$; in addition, we see $du = e^x dx$ in the integral. As x goes from 0 to $\ln 5$, $u = e^x + 4$ goes from $e^0 + 4 = 5$ to $e^{\ln 5} + 4 = 9$:

$$\begin{aligned} \int_0^{\ln 5} \frac{3e^x}{\sqrt{e^x + 4}} dx &= \int_5^9 \frac{3}{\sqrt{u}} du \\ &= \int_5^9 3u^{-1/2} du \\ &= 6u^{1/2} \Big|_5^9 \\ &= \boxed{18 - 6\sqrt{5}} \end{aligned}$$

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Reflect Back.

In which of the following integrals could you make a substitution to turn the integral into a constant multiple of $\int \sqrt{u} du$? What substitution would that be?

Example: Substituting $u = \sin x$ in $\int 2\sqrt{\sin x} \cdot \cos x dx$ gives $\int 2\sqrt{u} du$.

(a) $\int \sqrt{\tan x} dx$

Solution

No such substitution exists. (You might try to substitute $u = \tan x$, but $du = \sec^2 x dx$ is not present.)

(b) $\int \sqrt{\tan x} \cdot \sec^2 x dx$

Solution

Substituting $u = \tan x$ gives $\int \sqrt{\tan x} \cdot \sec^2 x dx = \int \sqrt{u} du$.

(c) $\int e^x \sqrt{e^{2x} + 1} dx$

Solution

No such substitution exists. (You might try to substitute $u = e^{2x} + 1$, but then $du = 2e^{2x} dx$.)

(d) $\int e^{2x} \sqrt{e^{2x} + 1} dx$

Solution

Substituting $u = e^{2x} + 1$ gives $\int e^{2x} \sqrt{e^{2x} + 1} \, dx = \int \frac{1}{2} \sqrt{u} \, du$.

(e) $\int \sqrt{3x+1} \, dx$

Solution

Substituting $u = 3x + 1$ gives $\int \sqrt{3x+1} \, dx = \int \frac{1}{3} \sqrt{u} \, du$.